

Prototype Lab Report – The Smart Water Bottle

ME 2900

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I. Introduction

The Smart Water Bottle prototype can pump water with variable power while sensing and displaying the temperature and level of the water. This prototype is intended to help the elderly and disabled members of society easily drink water again. Some users have difficulty opening bottle caps on typical water bottles. Refillable water bottles are often too heavy for some users to tip over to drink. Even sipping out of a straw can be too much work. To make drinking water simple for the user, all facets of drinking water will be automated. Water level and temperature sensors allow the user to see how much water and its temperature is in real time. These factors are displayed on the LCD screen and are to be monitored. The DC water pump removes the need to sip or tip the bottle. As a final product, the Arduino circuit would be reduced to a microchip and implanted into the removable base of a 64 OZ polypropylene water jug. Attached at the base powering the motor, circuits, and screen, will be a rechargeable 9-volt lithium battery. The bottle will glow blue due to a blue light installed to sanitize the water within the jug.

II. Discussion

❖ Idea Development Process

Once I had determined that the problem I wanted to solve was to help, the elderly, and disabled drink water, my process was straightforward. The goal of this prototype was to not only make drinking water easier, but better. But first, what would make it easy? Automating the process would make it easy. To simulate sipping water, I would need a water pump. Then I thought, "What would make it better?" I no longer thought of this prototype as just drinking water, this prototype was going to be a "Smart Water Bottle." From there, my ideas shifted to "What would a smart water bottle be capable of?" I then came up with:

- Reading the Temperature
- monitor the water level
- Track how much water, has been drunk.
- All metrics should be posted to an LCD Screen
- Water Pump should have variable power
- Water Pump should Be controlled by a Joystick

Despite being my first solution to this problem I did not come up with other prototype idea solutions. The brainstorming session produced a lot of good ideas that I wanted to pursue right off the bat.

❖ Other Solutions Considered

Realizing that there were going to be many inputs in my initial design, I figured that I would need two Shift registers daisy-chained together, to conserve Arduino input space. I was led away from this idea after one of the TAs had shown me that I was not implementing enough inputs to warrant a second shift register. I did not like the idea of having leftover water in the piping once the user had activated the water pump. I was hoping this could have been avoided by reversing the water pump but that would've been impossible with my current pump. It would have required a separate mechanical system to pump the leftover water back into the water, which would have been time-consuming, expensive, and impractical.

❖ Discussion of Prototype Operation/Features

The Smart Water Bottle implements several components into its design to operate as intended. DC water pump, water temperature sensor, water level sensor, joystick, and LCD. Once the joystick is moved, an analog signal is sent to the Arduino, which is then transformed into a number from 1 to 100 (depending on how far it is moved). This number will determine the percentage of power that will then be supplied to the DC Water pump. This accounts for the variable strength of the water current to the user.

Three variables are being transmitted and displayed on the LCD screen at all times: the joystick variable, water level variable, and water temperature variable. In the background, however, there is a condition where the joystick will not activate the water pump so long as the water level is below a certain threshold. This mitigates damage to the water pump from running dry. To achieve these features I established six user-defined functions to better organize my code. The six are as follows: Water temperature, water level, joystick, LCD screen, pump on, pump off. As the names suggest, they are all subroutines, to activate and control the different components.



Water temp sensor (Top left)

Water Temperature Sensor (Right)

DC Water Pump (Bottom Left)

LCD Screen displaying power to water pump, level of water, and temperature of water

Joystick to control water pump

❖ Difficulties

Creating conditional statements for the activation of the water pump via the joystick, and water level sensor was particularly difficult. The idea behind the water bottle is that the water pump won't activate until the joystick is activated, and the water level is above a certain threshold. Due to programming issues, and the finicky nature of the water level sensors, making this a seamless product was very difficult.

❖ Interesting/Successful Feature

Once it finally stopped when the water level had gotten low, and the water level metric on the LCD screen dropped accordingly, I was proud. This portion of the prototype was particularly challenging and when I finally got the code correct for the components to work together, I was happy.

❖ Changes/Improvements

I would have liked to invest more time into making it smaller. I could have achieved this by soldering the board together. I would have also bought a longer water level sensor as it is relatively short for what I am trying to achieve. Making it smaller and getting a longer water level sensor would've made producing a full-scale Smart Water bottle possible.

III. Conclusion

The purpose of this lab was to advance our engineering ability to solve problems with a novel solution. I was tasked with finding a solution to a problem that I saw in my personal life, and I had evaluated and created just that. This project was challenging, but rewarding in the potential of what this prototype could be should I go forward with it. I am grateful to Professor Eichaker, as well as the GTA's and TA's who assisted me in turning this idea and making it the real thing.

Appendix A: Arduino Sketches

```
//DC Water Pump
const int WaterPump = 6; //Motor on Digital Pin 6
int Power = 0;

//Water Temperature Sensor
#include <DallasTemperature.h>
#include <OneWire.h>
#define ONE_WIRE_BUS 2 //This is the digital pin number that
reads the input
OneWire oneWire(ONE_WIRE_BUS); //Use a 5K resistor as a pull-up
(bridging 5v and data)
DallasTemperature sensors(&oneWire); //red wire (5v) black wire (Gnd) yellow
wire (Data)
int TempC = 0;
int TempF = 0;

//Water Level Sensor
#define POWER_PIN 9
#define SIGNAL_PIN A5
int Level = 0; // variable to store the sensor value
int Level0 = 0;
int Signal = 0;

//Joystick
#include <ezButton.h>
#define VRY_PIN A0 // Arduino pin connected to VRY pin
int yValue = 0;

//LCD Screen
#include <LiquidCrystal_74HC595.h>
LiquidCrystal_74HC595 lcd(11, 13, 10, 1, 3, 4, 5, 6, 7);

void setup() {
    Serial.begin(9600);

    //Water Level Sensor
    pinMode(POWER_PIN, OUTPUT); // configure D9 pin as an OUTPUT
    digitalWrite(POWER_PIN, LOW); // turn the sensor OFF

    //Water Temperature Sensor
    sensors.begin();
    //Serial.print("Found devices: ");
    //Serial.println(sensors.getDeviceCount());
```

```

    //DC Water Pump
    pinMode(WaterPump, OUTPUT);
}

void loop() {
    WaterTemp();
    WaterLevel();
    yValue = Joystick();
    LCDScreen();

    if (yValue > 1 && Level > 25) { //Joy stick must be moved, and waterlevel
is greater than 25% Full
        PumpON();
    } else {
        PumpOFF();
    }
}

void WaterTemp() {
    sensors.requestTemperatures();
    //Serial.println("done");
    TempC = (sensors.getTempCByIndex(0)); //Retrieving temperature in degrees
celcius
    TempF = (TempC * 1.8) + 32;           //Converting to degrees Fahrenheit
}

void WaterLevel() {
    digitalWrite(POWER_PIN, HIGH); // turn the sensor ON
    delay(10);                      // wait 10 milliseconds
    //Serial.println(analogRead(SIGNAL_PIN));
    Signal = constrain(analogRead(SIGNAL_PIN), 150, 220);
    Level = map(Signal, 150, 220, 0, 100); // read the analog value from
sensor
    //Level = map(Level0, 76, 84, 0, 100); // stretching the variable to fine
tune
    digitalWrite(POWER_PIN, LOW); // turn the sensor OFF
    //Serial.println(Level);
}

int Joystick() {
    int yValue = map(abs(map(analogRead(VRY_PIN), 0, 1023, -512, 512)), 0, 512,
0, 100); // Converting analog input to absolute power percentage
    // Serial.println(yValue);
}

```

```

    delay(10);
    return yValue;
}

void LCDScreen() {
    lcd.begin(20, 4);

    //percent Power supplied to the Water Pump
    lcd.setCursor(0, 0); //Top Left Placement
    lcd.print(yValue);
    lcd.setCursor(3, 0);
    lcd.print("%Power");

    //Temperature of Water in Degrees Fahrenheit
    lcd.setCursor(3, 1); //Bottom Middle Placement
    lcd.print(TempF);
    lcd.setCursor(5, 1);
    lcd.print(" Degrees");

    lcd.setCursor(9, 0); // Top Right Placement
    lcd.print(Level);
    lcd.setCursor(12, 0);
    lcd.print("%H2O");
}

void PumpON() {
    Power = map(yValue, 0, 100, 0, 255);
    analogWrite(WaterPump, Power);
}

void PumpOFF() {
    digitalWrite(WaterPump, LOW);
}

```

Appendix B: CAD and Schematics



